



AGITATED NUTSCHE FILTER & DRYER

About **AMAR**

Serving Industry Since 1974

New 50000 sqft state-of-the-art manufacturing facility

Pioneer & largest manufacturer of lab scale ANFD in India

Inline with government's MAKE IN INDIA campaign

Unmatched quality & safety standards

ISO, CE-PED, ASME U, CSA, UL, Ex-Proof, ATEX certification

Prompt & efficient after sales service

Highly skilled & trained team of more than 200 personnel



Rotation

- High torque bidirectional motor
- RPM controller & current indicator

Manual agitator up / down

- Effortless up / down adjustment of agitator for drying

Agitator mechanical seal

- Vacuum and pressure rated

Agitator system

- Easily removable
- Adjustable height
- Provides smoothing & plowing in CW and CCW respectively

Jacketted vessel, lid & filter basket

- Heating / Cooling jacket for drying and cooling of cake
- Easily removable basket for dry or wet cake collection

Standard nozzles

- Slurry inlet / CIP / N₂ Purging, Vent
- Light & sight glass
- Compound pressure gauge

Vacuum / filtrate outlet nozzle

- To connect to vacuum pump through vacuum trap during contact & convective drying in parallel



AGITATED NUTSCHE FILTER AND DRYER



SALIENT FEATURES

- 0.005 m² to 0.20 m² filtration area
- 1 ltr to 100 ltr slurry volume
- Design pressure: -1 to 4 bar
- Design Temperature: - 20 to 150° C
- Materials: SS316, Hastelloy C, Inconel etc.
- Removable filter basket / disc size from 5 µm to 20 µm
- Transport & separation process takes place in single unit

- Reduced product exposure as all operation takes place in single unit
- Real-time model-predictive process control for the comparison of different agitation regimes & different operating conditions such as vacuum, pressure, nitrogen flow, jacket temperature & cake volume, together with comparison of different solvents
- Vacuum contact drying & through-circulation convective drying possible

APPLICATIONS

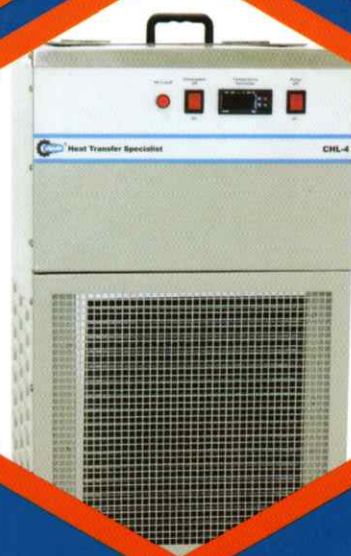
- Pharmaceutical Intermediates
- Food, flavors & fragrance industry
- Edible oil deodorization & deacidification
- Hazardous materials filtration
- Moisture removal & solvent recovery

- Precious metal & other catalyst recovery
- Fine & speciality chemicals
- Polymer devolatilization
- Microsphere separation & purification

STANDARD SPECIFICATIONS

Model*	Filter area	Slurry volume	Cake volume
ANFD 1	0.005 m ²	1 ltr	500 ml
ANFD 2	0.008 m ²	2 ltr	800 ml
ANFD 5	0.02 m ²	5 ltr	2 ltr
ANFD 10	0.05 m ²	10 ltr	5 ltr
ANFD 25	0.07 m ²	25 ltr	8 ltr
ANFD 50	0.13 m ²	50 ltr	18 ltr
ANFD 100	0.20 m ²	100 ltr	32 ltr

*Numeric value indicates slurry volume



HEATING COOLING CIRCULATORS

About **AMAR**

Serving the industry since 1974.

New 50000 sqft state-of-the-art manufacturing facility

Largest manufacturer of heating cooling circulators in India.

Inline with government's MAKE IN INDIA campaign.

An ISO 9001-2015 certified company.

Unmatched quality & safety standards

Manufacturing on CNC/VMC & automated machines

Prompt & efficient after sales service

Highly skilled & trained team of more than 200 personal

Single fluid closed loop system from -90°C to $+250^{\circ}\text{C}$

Heating bath circulators from ambient to $+200^{\circ}\text{C}$

Heating cooling bath circulators from -70°C to $+175^{\circ}\text{C}$

High temperature circulators from ambient to $+350^{\circ}\text{C}$

Chillers up to -30°C



SALIENT FEATURES

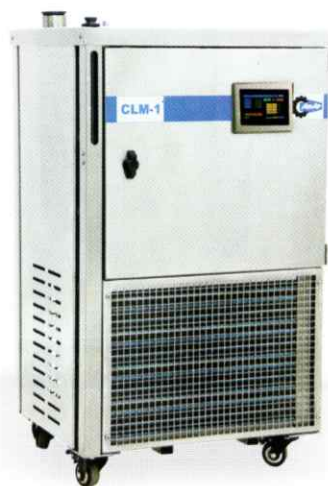
- High performance & energy efficient systems.
- Available for reactor volumes from 10 ml to 3000 ltr.
- Powerful pump for heat transfer fluid circulation.
- High pressure booster pump for flow reactors.
- Custom made systems for specific applications.
- Accurate & precise temperature control.
- Touch screen models.
- Eco-friendly & compact systems.
- Ex-proof systems for hazardous area.

APPLICATIONS

- Metal, GLR & Glass jacketed reactors (10 ml to 3000 ltr).
- High pressure autoclaves (100 ml to 2000 ltr).
- Micro & flow reactors.
- Rotary evaporators.
- Heat exchangers.
- Supercritical fluid extraction systems.
- Any other heat transfer applications.

In R&D centres, pilot plants & manufacturing facilities of fine & specialty chemicals, bulk drug pharmaceuticals, dyes, intermediates, paints, oils, agrochemical, oil & gas etc. industries.

SINGLE FLUID CLOSED LOOP SYSTEM



SALIENT FEATURES

- Temperature range: -90°C to +250°C
- Temperature controller type: PID cascade control
- Temperature display: Touch screen TFT display (7")
- Temperature control type: Jacket & / or process temperature control selectable
- Display resolution: 0.01°C
- Temperature sensors: RTD PT 100
- Temperature control accuracy: $\pm 0.5^\circ\text{C}$
- Heating: Electrical heating
- Cooling: Refrigeration system (CFC free refrigerants)
- Complete SS-304 Construction
- RS 485 communication port with SCADA software
- Multistep ramp soak programmable controller
- Data logging / recording optional
- Mounting on suitable caster wheels
- Suitable chillers for water cooled machines

STANDARD SAFETY FEATURES

- System over temperature cut-off: Independent
- Compressor over pressure cut-off
- Heater, pump & compressor overload cut-off
- Compressor over temperature cut-off
- Safety interlock pump no flow
- Compressor oil pressure difference cut-off

CLM SERIES (-90°C TO +200°C) - FOR LAB APPLICATIONS

Model	Temperature Range °C	Heating KW	Cooling power kW			Pump Flow Max (lpm)*	Max Pump Pressure (bar)	Dimensions mm (W x D x H)	End Connection
			0°C	-20°C	-30°C				
CLM-1	-30 to +200	1.5	0.6	0.3	0.1	18~20	0.4	475 x 400 x 800	M.24 Male
CLM-2	-35 to +200	2.5	0.8	0.5	0.25	30~35	0.4	420 x 520 x 875	M.24 Male
CLM-3	-35 to +200	3.6	1.2	0.7	0.3	30~35	0.4	450 x 600 x 900	M.24 Male
CLM-4W	-35 to +200	4.5	2.5	1.3	0.6	40~45	0.7	540 x 650 x 1250	M.42 Male
			0°C	-40°C	-60°C				
CLL-1	-75 to +180	1.5	0.6	0.6	0.3	30~35	0.4	600 x 650 x 1400	M.24 Male
CLL-2	-75 to +180	3	1.5	1.4	1.2	30~35	0.4	600 x 650 x 1400	M.24 Male
CLL-3W	-75 to +180	4.5	2.2	2.0	1.4	40~45	0.7	700 x 750 x 1500	M.42 Male
CLL-4W	-90 to +180	4.5	2.2	2.0	1.4	40~45	0.7	700 x 750 x 1500	M.42 Male

Note:

- The given cooling power is at 30°C ambient temperature. Increase in ambient temperature will affect performance of machine
- Suffix "W" in model is for water cooled machines
- The temperature range mentioned above is at outlet of the machine hence temperature inside the process / load may vary due to various factors like process design, distance, head, insulation etc.
- *Pump flow rates are at atmospheric pressure
- Since development is continuous process, the above specifications are subjected to change without prior notice.



SINGLE FLUID CLOSED LOOP SYSTEM



CPM SERIES (-60°C TO +200°C) - FOR PILOT APPLICATIONS

Model	Temperature Range °C	Heating KW	Cooling power kW			Pump Flow Max (lpm)*	Max Pump Pressure (bar)	Dimensions mm (W x D x H)	End Connection
			0°C	-20°C	-30°C				
CPM-1W	-35 to +200	6	4	2.2	1.3	50~55	1.0	560 x 650 x 1300	M.42 Male
CPM-2W	-35 to +200	9	7	3.5	1.8	50~55	1.0	600 x 690 x 1300	M.42 Male
CPM-3W	-30 to +200	12	12	6	3	80~90	1.5	700 x 750 x 1500	M.42 Male
CPM-4W	-30 to +200	18	15	7	3.5	80~90	1.5	750 x 790 x 1500	M.42 Male
			0°C	-40°C	-60°C				
CPL-1W	-60 to +200	6	6.5	3.2	1	50~55	1.0	750 x 790 x 1500	M.42 Male
CPL-2W	-60 to +200	9	12	6	1.5	80~90	1.5	750 x 790 x 1500	M.42 Male

CIM SERIES (-30°C TO +170°C) - FOR INDUSTRIAL APPLICATIONS

Model	Temperature Range °C	Heating KW	Cooling power kW			Pump Flow Max (lpm)*	Max Pump Pressure (bar)	Dimensions mm (W x D x H)	End Connection
			0°C	-20°C	-30°C				
CIM-1W	-30 to +170	21	25	11	8	100	1.5	1200x950x1600	M.42 Male
CIM-2W	-30 to +170	33	45	21	12	120	2	1400x950x1800	M.42 Male
CIM-3W	-30 to +170	45	70	30	17	120	2	1800x1000x1600	M.42 Male
CIM-4W	-30 to +170	66	90	45	24	120	2	2000x1200x2000	M.42 Male

Note:

- Suffix "W" in model is for water cooled machine
- The given cooling power is at 35°C ambient temperature. Increase in ambient temperature will affect performance of machine
- The temperature range mentioned above is at outlet of the machine hence temperature inside the process / load may vary due to various factors like process design, distance, head, insulation etc.
- *Pump flow rates are at atmospheric pressure
- Custom built models on request for larger capacity
- Since development is continuous process, the above specifications are subjected to change without prior notice.

HEAT TRANSFER FLUIDS

HTF Grade	Working temp. range
D12	-60 to +200°C
XLT	-80 to +180°C

HEATING BATH CIRCULATORS



STANDARD FEATURES

- Temperature range: ambient to +350°C
- Temperature controller type: PID
- External load and bath temperature controller
- Display resolution: 0.1°C
- Temperature sensors: RTD PT 100
- Temperature control accuracy: $\pm 0.5^\circ\text{C}$
- Heating: Electrical heating
- Cooling: Through internal cooling coil
- Complete SS-304 construction
- RS 485 communication port with SCADA software
- Multistep ramp soak programmable controller
- Optional: N₂ purging facility to prevent fuming
- Mounting on suitable caster wheels



Touch control panel
for HTC series



Flame proof control panel
for HTC series

HB SERIES (AMBIENT TO 200°C) - HEATING BATH

Model	Temperature Range	Heating kW	Pump Flow Max (lpm)*	Max Pump Pressure (bar)	Dimensions mm (W x D x H)	End Connection
HB-1	Amb. to +200	1	14~16	0.4	300 x 500 x 450	M.16 Male
HB-2	Amb. to +200	2	14~16	0.4	430 x 560 x 550	M.16 Male
HB-3	Amb. to +200	3	30~35	0.7	510 x 620 x 600	M.24 Male

HTC SERIES (AMBIENT TO 350°C) - HIGH TEMPERATURE CIRCULATOR

Model	Temperature Range	Heating kW	Pump Flow Max (lpm)*	Max Pump Pressure (bar)	Dimensions mm (W x D x H)	End Connection
HTC-1	Amb. to +350	3	30~35	0.7	410 x 410 x 700	M.24 Male
HTC-2	Amb. to +350	6	30~35	0.7	600 x 600 x 900	M.24 Male
HTC-3	Amb. to +350	12	50~55	1	900 x 900 x 1100	M.42 Male

- Control panel dimensions: 300 x 300 x 300 mm

Note:

- The temperature range mentioned above is at outlet of the machine hence temperature inside the process / load may vary due to various factors like process design, distance, head, insulation etc.
- In HTC series, machines with higher heating capacity also available on request
- In HTC series, flame proof options are available on request
- *Pump flow rates are at atmospheric pressure
- Since development is continuous process, the above specifications are subjected to change without prior notice.

HEAT TRANSFER FLUIDS FOR HB SERIES

HTF Grade	Working temp. range
S 50	Amb. to +200°C

HEAT TRANSFER FLUIDS FOR HTC SERIES

HTF Grade	Working temp. range
T 55	Amb. to +320°C
T 66	Amb. to +350°C



HEATING COOLING BATH CIRCULATORS



STANDARD FEATURES

- Temperature range: -70°C to +175°C
- Temperature controller type: PID
- External load and bath temperature controller
- Display resolution: 0.1°C
- Temperature sensors: RTD PT 100
- Temperature control accuracy: $\pm 0.5^\circ\text{C}$
- Heating: Electrical heating
- Cooling: Refrigeration System (CFC free refrigerants)
- Complete SS-304 Construction
- RS 485 communication port with SCADA software
- Multistep ramp soak programmable controller
- N2 purging facility to prevent fuming
- Mounting on suitable caster wheels

HCB SERIES (-25°C TO +175°C) - HEATING COOLING BATH

Model	Temperature Range °C	Heating KW	Cooling power kW			Pump Flow Max (lpm)*	Max Pump Pressure (bar)	Dimensions mm (W x D x H)	End Connection
			0°C	-20°C	-25°C				
HCB-1	-25 to +175	1	0.4	0.2	0.05	14~16	0.4	300 x 500 x 650	M.16 Male
HCB-2	-25 to +175	2	1.1	0.7	0.3	14~16	0.4	420 x 520 x 1000	M.16 Male
HCB-3	-25 to +175	3	1.5	1.0	0.45	30~35	0.7	510 x 620 x 1050	M.24 Male

HCL SERIES (-70°C TO +100°C) - HEATING COOLING CRYO BATH

Model	Temperature Range °C	Heating KW	Cooling power kW			Pump Flow Max (lpm)*	Max Pump Pressure (bar)	Dimensions mm (W x D x H)	End Connection
			-20°C	-40°C	-60°C				
HCL-1	-70 to +100	1	0.4	0.3	0.2	14~16	0.4	610 x 700 x 1150	M.16 Male
HCL-2	-70 to +100	2	1.1	0.8	0.4	14~16	0.4	610 x 700 x 1150	M.16 Male
HCL-3	-70 to +100	3	1.5	1.2	0.6	30~35	0.7	700 x 700 x 1300	M.24 Male

Note:

- The given cooling power is at 30°C ambient temperature. Increase in ambient temperature will affect performance of machine
- The temperature range mentioned above is at outlet of the machine hence temperature inside the process / load may vary due to various factors like process design, distance, head, insulation etc.
- *Pump flow rates are at atmospheric pressure
- Since development is continuous process, the above specifications are subjected to change without prior notice.

HEAT TRANSFER FLUIDS

HTF Grade	Working temp. range
S 5	-70 to +100°C
S 20	-25 to +175°C

CHILLERS FOR LABORATORY APPLICATIONS



CHL SERIES (AMBIENT TO -15°C)

Model	Temperature Range °C	Cooling power in kW @ (Bath Fluid IPA)		Pump Flow (lpm)*	Dimensions mm (W x D x H)	End Connection
		0°C	-10°C			
CHL-1	Ambient to -15	0.35	0.15	12~14	330 x 370 x 550	3/8" Hose Nipple
CHL-2	Ambient to -15	0.7	0.3	12~14	360 x 460 x 600	1/2" Hose Nipple
CHL-3	Ambient to -15	1.2	0.6	25~30	550 x 650 x 700	1/2" Hose Nipple
CHL-4	Ambient to -15	2.4	1.5	30~35	450 x 650 x 1150	3/4" Hose Nipple
CHL-5	Ambient to -15	4.5	2.5	30~35	700 x 700 x 1200	3/4" Hose Nipple
CHL-6	Ambient to -15	10	6	55~60	1300 x 800 x 1600	1" Hose Nipple
CHL-7	Ambient to -15	15	8	70 ~ 75	1500 x 950 x 1700	1" Hose Nipple
CHL-8	Ambient to -15	20	12	85~90	1650 x 1100 x 1600	1" Hose Nipple

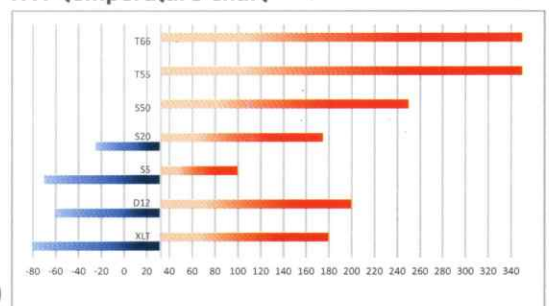
Note:

- The given cooling power is at 30°C ambient temperature. Increase in ambient temperature will affect performance of machine
- The temperature range mentioned above is at outlet of the machine hence temperature inside the process / load may vary due to various factors like process design, distance, head, insulation etc.
- *Pump flow rates are at atmospheric pressure
- Since development is continuous process, the above specifications are subjected to change without prior notice.

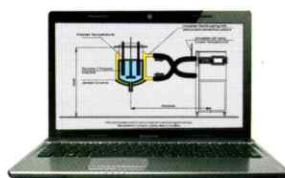
OPTIONAL ACCESSORIES

- Synthetic eat transfer fluid for wide temperature range (non toxic & non flammable)
- SCADA software for PC operated / programmable heating or cooling machines with online & historical data Logging.
- Flexible SS hose pipes with insulation for fluid circulation
- Purge panel for complete unit to use in Ex-proof area
- Flame proof panel for remote set point

HTF temperature chart



(a)



(b)



(c)



(d)



(e)



APPLICATIONS USING VARIOUS MODELS

CLM for AMaR-3 &
AMaR-4P flow reactors



Heating cooling bath for high
pressure polystest reactor system



Heating cooling bath & chiller
for supercritical fluid circulation



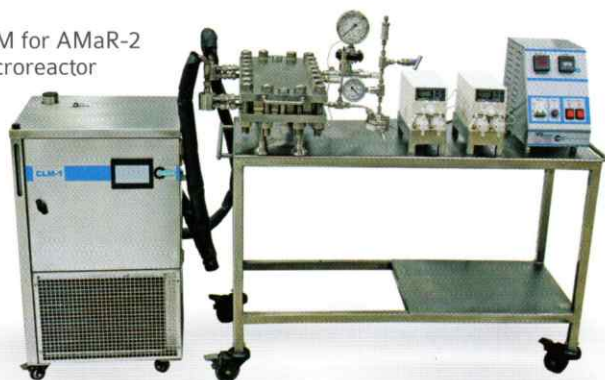
Heating cooling bath
for microreactor



APPLICATIONS USING VARIOUS MODELS



CLM for AMaR-2 microreactor



Heating cooling bath for microreactor



Heating cooling bath for glass reactor



CLL-2 with 50 ltr reactor



Heating cooling bath for flow reactor





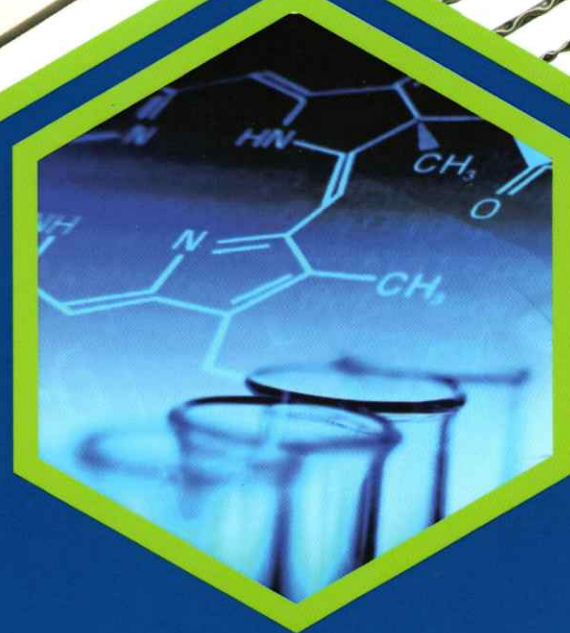
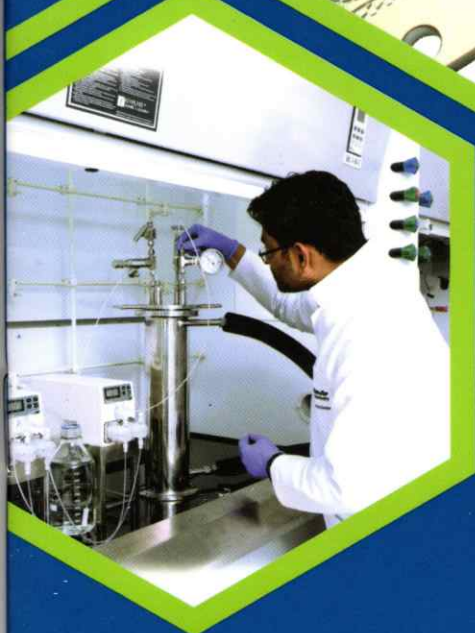
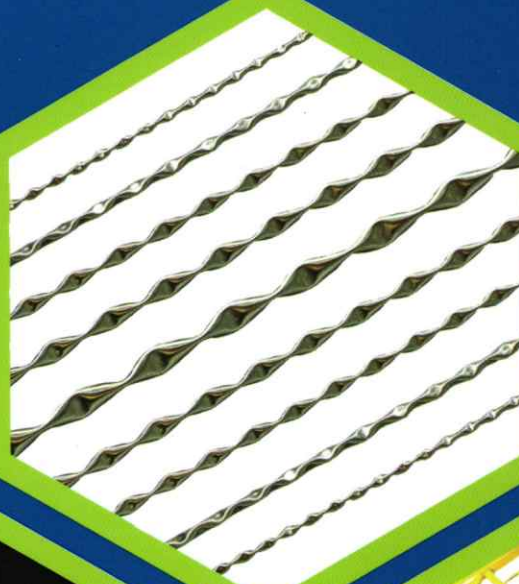
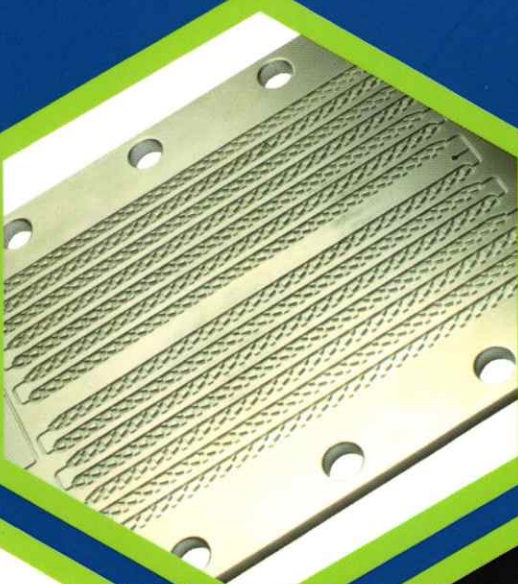
SPECIFICATIONS FOR ALL MODELS

Model	Temperature range °C	Heating power in Kw	Cooling power in Kw @					Bath volume filling/total	Bath opening/depth mm LxW/D		Display resolution	Temperature stability	Display type	Pump flow lpm	Maximum pressure bar	HTF end connection	Dimensions mm	Supply voltage	Maximum current	Machine version	Ambient temperature	Cooling water flow and pressure
			Cooling power in Kw @																			
			0°C	-20°C	-30°C	-40°C	-60°C															
CLM-1	-30...+200	1.5	0.6	0.3	0.1	—	—	—	—	0.01	±0.5	Touch	18~20	0.3	M-24	475x400x800	1-ph 230 VAC	10A	Air cooled	10°C~35°C	—	
CLM-2	-35...+200	2.5	0.8	0.5	0.25	—	—	—	—	0.01	±0.5	Touch	30~35	0.4	M-24	420x520x875	3-ph 420 VAC	10A	Air cooled	10°C~35°C	—	
CLM-3	-35...+200	3.6	1.2	0.7	0.3	—	—	—	—	0.01	±0.5	Touch	30~35	0.4	M-24	450x600x900	3-ph 420 VAC	12A	Air cooled	10°C~35°C	—	
CLM-4	-35...+200	4.5	2.5	1.3	0.6	—	—	—	—	0.01	±0.5	Touch	40~45	0.7	M-42	540x650x1250	3-ph 420 VAC	15A	Air cooled	10°C~35°C	250 lph/3 bar	
CLL-1	-75...+180	1.5	0.6	0.6	0.6	0.3	—	—	—	0.01	±0.5	Touch	30~35	0.4	M-24	600x650x1400	3-ph 420 VAC	15A	Air cooled	10°C~35°C	—	
CLL-2	-75...+180	3	1.5	1.5	1.5	1.4	1.2	—	—	0.01	±0.5	Touch	30~35	0.4	M-24	600x650x1400	3-ph 420 VAC	18A	Air cooled	10°C~35°C	—	
CLL-3	-75...+180	4.5	2.2	2.2	2.1	2	1.4	—	—	0.01	±0.5	Touch	40~45	0.7	M-42	700x750x1500	3-ph 420 VAC	20A	Water cooled	10°C~35°C	250 lph/3 bar	
CLL-4	-90...+180	4.5	2.2	2.2	2.1	2	1.4	—	—	0.01	±0.5	Touch	40~45	0.7	M-42	700x750x1501	3-ph 420 VAC	20A	Water cooled	10°C~35°C	250 lph/3 bar	
CPM-1	-35...+200	6	4	2.2	1.3	—	—	—	—	0.01	±0.5	Touch	50~55	1	M-42	560x650x1300	3-ph 420 VAC	15A	Water cooled	10°C~35°C	400 lph/3 bar	
CPM-2	-35...+200	9	7	3.5	1.8	—	—	—	—	0.01	±0.5	Touch	50~55	1	M-42	600x690x1300	3-ph 420 VAC	35A	Water cooled	10°C~35°C	700 lph/3 bar	
CPM-3	-30...+200	12	12	6	3	—	—	—	—	0.01	±0.5	Touch	80~90	1.5	M-42	700x750x1500	3-ph 420 VAC	50A	Water cooled	10°C~35°C	1200 lph/3 bar	
CPM-4	-30...+200	18	15	7	3.5	—	—	—	—	0.01	±0.5	Touch	80~90	1.5	M-42	750x790x1500	3-ph 420 VAC	50A	Water cooled	10°C~35°C	1500 lph/3 bar	
CPL-1	-60...+200	6	6.5	6.5	6.5	3.2	1	—	—	0.01	±0.5	Touch	50~55	1	M-42	750x790x1500	3-ph 420 VAC	40A	Water cooled	10°C~35°C	650 lph/3 bar	
CPL-2	-60...+200	9	12	12	12	6	1.5	—	—	0.01	±0.5	Touch	80~90	1.5	M-42	750x790x1500	3-ph 420 VAC	50A	Water cooled	10°C~35°C	1200 lph/3 bar	
CIM-1	-30...+170	21	25	11	8	—	—	—	—	0.01	±1	Touch	100	1.5	M-42	1200x950x1600	3-ph 420 VAC	65A	Water cooled	10°C~35°C	2500 lph/3 bar	
CIM-2	-30...+170	33	45	21	12	—	—	—	—	0.01	±1	Touch	120	2	M-42	1400x950x1800	3-ph 420 VAC	75A	Water cooled	10°C~35°C	4500 lph/3 bar	
CIM-3	-30...+170	45	70	30	17	—	—	—	—	0.01	±1	Touch	120	2	M-42	1800x1000x1600	3-ph 420 VAC	95A	Water cooled	10°C~35°C	7000 lph/3 bar	
CIM-4	-30...+170	66	90	45	24	—	—	—	—	0.01	±1	Touch	120	2	M-42	2000x1200x2000	3-ph 420 VAC	110A	Water cooled	10°C~35°C	9000 lph/3 bar	
HCB-1	-25...+175	1	0.4	0.2	0.05	—	—	8/10	150x130/140	0.1	±0.5	LED/LCD	14~16	0.4	M-16	300x500x650	1-ph 230 VAC	8A	Water cooled	10°C~35°C	—	
HCB-2	-25...+175	2	1.1	0.7	0.3	—	—	12/16	155x100/150	0.1	±0.5	LED/LCD	14~16	0.4	M-16	420x560x1000	1-ph 230 VAC	12A	Air cooled	10°C~30°C	—	
HCB-3	-25...+175	3	1.5	1	0.45	—	—	21/27	200x140/170	0.1	±0.5	LED/LCD	30~35	0.7	M-24	510x620x1050	3-ph 420 VAC	8A	Air cooled	10°C~30°C	—	
HCL-1	-70...+100	1	—	0.4	—	0.3	0.2	8/10	150x130/140	0.1	±0.5	LED/LCD	14~16	0.4	M-16	610x700x1150	1-ph 230 VAC	15A	Air cooled	10°C~30°C	—	
HCL-2	-70...+100	2	—	1.1	—	0.8	0.4	12/16	155x100/150	0.1	±0.5	LED/LCD	14~16	0.4	M-16	610x700x1150	1-ph 230 VAC	20A	Air cooled	10°C~30°C	—	
HCL-3	-70...+100	3	—	1.5	—	1.4	0.6	21/27	200x140/170	0.1	±0.5	LED/LCD	30~35	0.7	M-24	700x700x1300	3-ph 420 VAC	15A	Air cooled	10°C~30°C	—	
HB-1	Amb...+200	1.5	Through internal cooling coil					8/10	120x140/150	0.1	±0.5	LED/LCD	14~16	0.4	M-16	300x500x450	1-ph 230 VAC	6A	—	10°C~30°C	—	
HB-2	Amb...+200	2	Through internal cooling coil					12/16	200x150/150	0.1	±0.5	LED/LCD	14~16	0.4	M-216	420x560x550	1-ph 230 VAC	10A	—	10°C~30°C	—	
HB-3	Amb...+200	3	Through internal cooling coil					21/27	200x150/170	0.1	±0.5	LED/LCD	30~35	0.7	M-24	510x620x600	3-ph 420 VAC	8A	—	10°C~30°C	—	
HTC-1	Amb...+350	3	Through internal cooling coil					—	—	0.1	±0.5	LED/LCD	30~35	0.7	M-24	410x410x700	1-ph 230 VAC	18A	—	10°C~30°C	—	
HTC-2	Amb...+350	6	Through internal cooling coil					—	—	0.1	±0.5	LED/LCD	30~35	0.7	M-24	600x600x900	3-ph 420 VAC	12A	—	10°C~30°C	—	
HTC-3	Amb...+350	12	Through internal cooling coil					—	—	0.1	±0.5	LED/LCD	50~55	1	M-42	900x900x1100	3-ph 420 VAC	22A	—	10°C~30°C	—	
			0°C -10°C																			
CHL-1	Amb...-15	—	0.35	0.15	—	—	—	—	—	0.1	±2	LED/LCD	12~14	0.3	M-24	330x370x550	1-ph 230 VAC	6A	Air cooled	10°C~30°C	—	
CHL-2	Amb...-15	—	0.7	0.3	—	—	—	—	—	0.1	±2	LED/LCD	12~14	0.3	M-24	360x460x600	1-ph 230 VAC	8A	Air cooled	10°C~30°C	—	
CHL-3	Amb...-15	—	1.2	0.6	—	—	—	—	—	0.1	±2	LED/LCD	25~30	0.5	M-24	550x650x700	1-ph 230 VAC	12A	Air cooled	10°C~30°C	—	
CHL-4	Amb...-15	—	2.4	1.5	—	—	—	—	—	0.1	±2	LED/LCD	30~35	0.6	M-24	450x650x1150	3-ph 420 VAC	6A	Air cooled	10°C~30°C	—	
CHL-5	Amb...-15	—	4.5	2.5	—	—	—	—	—	0.1	±2	LED/LCD	30~35	0.6	M-24	700x700x1200	3-ph 420 VAC	8A	Air cooled	10°C~30°C	—	
CHL-6	Amb...-15	—	10	6	—	—	—	—	—	0.1	±2	LED/LCD	55~60	1	M-24	1300x800x1600	3-ph 420 VAC	25A	Air cooled	10°C~30°C	—	
CHL-7	Amb...-15	—	15	8	—	—	—	—	—	0.1	±2	LED/LCD	55~60	1	M-24	1500x950x1700	3-ph 420 VAC	32A	Air cooled	10°C~30°C	—	
CHL-8	Amb...-15	—	20	12	—	—	—	—	—	0.1	±2	LED/LCD	80~90	1.5	M-24	1650x1100x1600	3-ph 420 VAC	45A	Air cooled	10°C~30°C	—	

OUR VALUED CLIENTELE



 Aarti Drugs Ltd.	 aether elementally innovative	 Alembic Touching Lives over 100 years	 Citul A member of Lalbhai Group	 ANUPAM RASAYAN INDIA LTD
 BARC भारतीय रसायन संस्थान भारत में सेवा के लिए	 Biocon	 CADILA PHARMACEUTICALS LIMITED	 Cipla Caring for life	 dishman
 Divis Divis Laboratories Ltd.	 DORF KET L	 Dr.Reddy's	 Eisai	 EVONIK INDUSTRIES
 FAIRCHEM A FAIRFAX COMPANY	 Firmenich for good, naturally	 FRESENIUS KABI caring for life	 EXCEL	 GE
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 ipca A dose of life	 JDMResearch	 ISRO	 Jay Chemicals Global Supplier of Fine and Specialty Chemicals	 Johnson Matthey
 LARSEN & TOUBRO	 LUPIN PHARMACEUTICALS, INC.	 MACLEODS	 MERCK	 Mylan
 MOMENTIVE performance materials	 NATCO	 JUBILANT ORGANOSYS ScienceActive	 Pfizer	 PIL Inspired by Science
 Piramal Healthcare knowledge action care	 POLYGENTA This Way	 PTI Polymer Technologies International	 SAI ADVANTIUM	 SANDOZ
 SANOFI	 SRF MAKING OUR NATION PROUD	 SCL Saurav Chemicals Ltd., Derabassi, Punjab - India.	 SRVATHI artificial intelligence in chemistry innovations...	 SUDARSHAN
 SUN PHARMACEUTICAL INDUSTRIES LTD.	 Syngene A Biocon company	 syngenta	 THEMIS MEDICARE	 Unimark Remedies Ltd. CREATING POSITIVE REACTIONS
 United Phosphorus Ltd	 USV	 valOrganics Pvt. Ltd.	 VINATI ORGANICS LIMITED	 VVF VVF Limited



CONTINUOUS FLOW TECHNOLOGY

SALIENT FEATURES

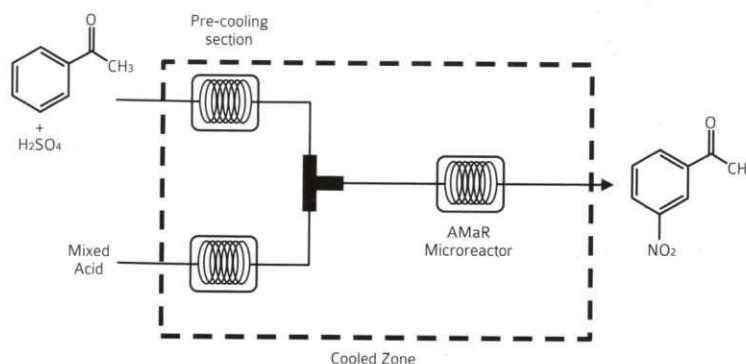
- Available in integrated microchannel & tubular metal construction for continuous flow process
- Reactors from 1 ml to 200 ltr
- 3D printed metal microreactors
- Very high heat and mass transfer coefficient
- Pressures up to 350 bar & temperatures up to 500°C
- SS316, Hastelloy, Inconel, Titanium etc. materials
- Suitable for various liquid-liquid, gas-liquid - multiphase reactions, and reactions involving solids
- Lab to plant scale turnkey solutions with pumps, utilities, pressure & flow control, safety device & controls in SCADA
- Easy scale-up from lab to production
- AMaR series of flow reactors are developed in technical collaboration with NCL, Pune

APPLICATIONS

AMaR series is ideal for carrying out a wide range of chemical reactions in the field of pharmaceuticals, fine and specialty chemicals in a continuous manner.

Suitable solutions for

- Homogeneous reactions: Neutralization, condensation, dehydration etc.
- Multiphase Reactions:
 - Gas-liquid reactions (G-L): oxidation, ozonolysis, halogenation, chlorination, bromination etc.
 - Liquid-liquid reactions (L-L): nitration, diazotization, azo coupling, transfer hydrogenation, sulfoxidation, amination, nitration of aromatic substrate, acylation, formylation, methylation, synthesis of nanomaterials, knoevenagel condensation, meerwein arylation, glycosylation hydrolysis, alkylation, sulfonation, sulfoxidation, synthesis of deuterated solvents, acetylation, oximation, cyclization etc.
 - G-L-S and L-S catalytic reactions: hydrogenation, pyrolysis, gasification, vapour phase reactions, etc.



GUIDELINES ON WHEN TO USE AMaR SERIES OF FLOW REACTORS

- **Inconsistent batch performance** due to high sensitivity towards process variables (temperature, pressure, concentration, time and mixing)
- **Process needs**
 - High heat transfer area
 - Efficient mixing
 - Rapid mass transfer
 - High interfacial area
- **Safety issues in**
 - Storing large inventories of hazardous reactants
 - Handling of hazardous and unstable chemicals
 - Generation of unstable intermediate by-products during the reaction
 - Runaway reactions
- **Process waste related issues**
 - Need for enhancing the selectivity close to theoretical value
 - Avoiding generation of waste that would need treatment
 - Need to avoid or reduce the downstream processing efforts for purification
- **Distributed production**
 - Produce only as per the local requirement/ consumption (location wise production)
 - Produce only as and when needed (avoid inventories)
- **Optimal utilization of facility**
 - A continuous flow manufacturing facility can also be used for production of multiple products with almost same peripherals
 - Space occupancy will be less
 - Achieve same or even better results than conventional batch operation



AMaR-1 LAB MICROREACTOR

SALIENT FEATURES

- Screening tool for continuous flow feasibility studies
- Suitable for reaction of miscible, immiscible liquids & gases
- Patented microchannel design
- Extremely effective mixing
- Very high heat transfer area
- Easy to clean - openable design
- Economical
- Ideal for fast & exothermic reactions

TECHNICAL SPECIFICATIONS

Volume	1 ml + 10 ml residence coil
Flow rate (lph)	Upto 40
Design pressure (bar)	50
Design temperature (°C)	-30 to 250 (with FFKM-Kalrez 'O' ring)
Material of construction	SS 316 / Hastelloy C- 276
Heat transfer area (m ² /m ³)	>2000



AMaR-2 MICROREACTOR

SALIENT FEATURES

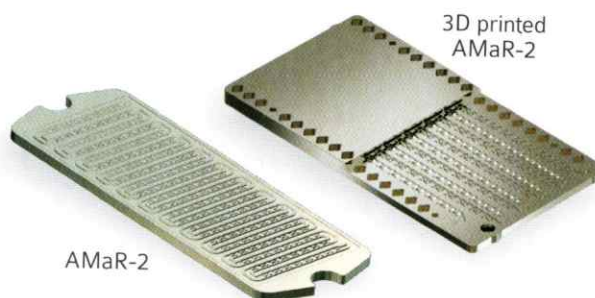
- Patented microchannel design with alternate reactor & utility plates
- Modular design for easy scale-up from lab to production
- Very high specific surface area for effective heat transfer
- Openable design for ease of maintenance
- Alternate 3D printed fused design for very high heat transfer coefficient

APPLICATIONS

AMaR-2 is suitable for fast and highly exothermic liquid-liquid reactions. Some proven applications include reactions like nitration, halogenations, sulfonation, diazotization, oxidation/reduction, etc.

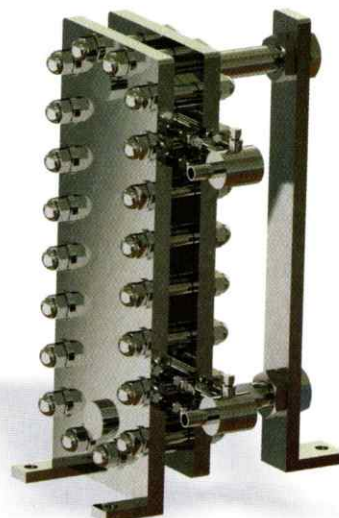
TECHNICAL SPECIFICATIONS

	Standard	Optional
Volumes	12 ml per plate	Upto 120 ml with 10 plates
	50 ml per plate	Upto 500 ml with 10 plates
	200 ml per plate	Upto 2000 ml with 10 plates
Flow rate (lph)	Upto 720	Upto 7200
Design pressure (bar)	20	50
Design temperature (°C)	-50 to 200	-50 to 350
Material of construction	SS 316	Hastelloy C 276
Heat transfer area (m ² /m ³)	500 - 2500	----



AMaR-2

3D printed
AMaR-2



AMaR-3: CONICAL FLOW REACTOR



SALIENT FEATURES

- Patented conical 3D design offering effective mixing
- High pressure capability as completely fused design
- Suitable for mass transfer limiting reactions
- Modular construction hence easily scalable

APPLICATIONS

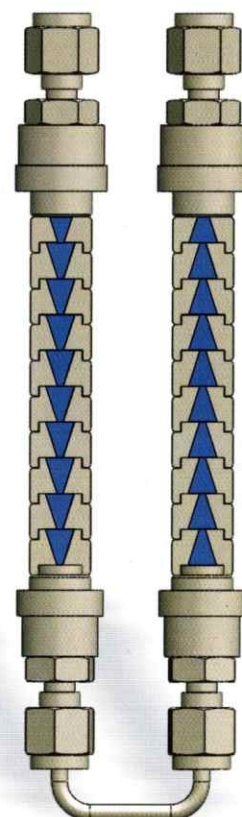
AMaR-3 is suitable for liquid-liquid immiscible reactions & gas-liquid reactions. Some proven applications include fast and exothermic reactions like halogenations, oxidation, condensation reactions, etc.

TECHNICAL SPECIFICATIONS

Volumes	1 ml to 200 ml
Flow rate (lph)	Upto 720
Design pressure (bar)	100
Design temperature (°C)	-50 to 350
Material of construction (MOC)	SS 316 / Hastelloy C 276
Heat transfer area (m ² /m ³)	500 - 2600



3D view of AMaR-3 assembly



Internal flow through candle

AMaR-4P: PINCHED TUBE FLOW REACTOR

SALIENT FEATURES

- Patented pinched-tube design for effective mixing
- Pilot to plant scale flow reactor with different size pinch tubes
- High heat transfer surface area due to pinching
- Metric tons production capacity
- Suitable for mixing of miscible, immiscible liquids & gases
- Most economical reactor in terms of cost / unit volume of reactor
- Effective for continuous L-L extraction

APPLICATIONS

AMaR-4P is a versatile reactor with applications ranging from miscible/immiscible liquid-liquid reactions, gas-liquid reactions, liquid-liquid extraction. They can be used in conjunction with AMaR-2 or AMaR-3 to provide the necessary residence time for desired conversion.

TECHNICAL SPECIFICATIONS

Volumes	100 ml to 200 ltr.
Flow rate (lph)	upto 10000
Design pressure (bar)	50
Design temperature (°C)	-50 to 350
Material of construction (MOC)	SS 316 / Hastelloy C 276
Heat transfer area (m ² /m ³)	200 - 1400





TUBULAR REACTOR & STATIC MIXERS



TECHNICAL SPECIFICATIONS

	Standard	Optional
Volumes	100 ml to 200 ltr.	Customized
Flow rate (lph)	up to 10000	-
Design pressure (bar)	100 bar	Up to 350
Design temperature (°C)	-50 to 350	-50 to 350
Material of construction (MOC)	SS 316	Hastelloy C

APPLICATIONS

- Optional static mixer elements for better mixing.
- Mixing of high viscosity fluids, gas-liquid & slurry applications.
- Dispersion of immiscible liquids.
- Mixing of acids, juices, oils etc.



PHOTO-FLO REACTOR

SALIENT FEATURES

- UV transparent chemically inert tubes
- Reactor volumes 12, 50, 200, 1000 & 2000 ml
- Flow rates up to 240 ltr/hr
- Pressures up to 18 bar & 150 °C temperature
- Efficient, uniform irradiation
- Precise control over exposure time
- For superior selectivity
- Safer operations compared to batch (bulk of solvent is away from lamp)
- Scale independent (capable of producing from few gm to kgs)
- Highly concentrated solution can be irradiated
- Photochemical flow reactor combines the advantages of continuous flow processing with the added benefits of ultraviolet light to enable photochemical reactions and a consistent distribution of UV light.

APPLICATIONS

- Photo chlorination
- Production of vitamin D
- Photo alkylation
- Artemisinin production (anti malarial drug)
- Production of e-caprolactame



FLOW REACTOR & SYSTEMS



GLASS MICROREACTOR



- Volume: 0.1 ml to 50 ml.
- Up to 15 bar & 300 °C
- Material: Glass, quartz, silicon, glass-silicon compounds.
- Integrated multilayer glass construction for mixing, reaction & heat transfer.

MULTI-FLO REACTOR



- Cost-effective starter set for proof-of-concept studies.
- Interchangeable reactor modules of AMaR-2, AMaR-3, AMaR-4, AMaR-4P with heating cooling.
- Up to 50 bar & -50°C to 350°C.
- Material: SS316, Hastelloy C.

ELECTROCHEMICAL FLOW REACTOR



- Volume: 5 ml to 3000 ltr.
- Design pressure: Up to 2 bar & 120°C.
- Wetted part: PTFE.
- Graphite, lead, copper, platinum, aluminium, iron, titanium, Mg, SS, Zn, Cu/Ni electrodes.

LIQUID LIQUID SEPARATOR



- Flow rate: Up to 12 lph.
- Design temperature: Up to 120°C.
- Wetted part: PTFE.
- Advanced membrane-based liquid-liquid separation.
- Highly efficient than gravity based separations, even for challenging emulsions.



SOLI-FLO REACTOR

SALIENT FEATURES

- Homogeneous and heterogeneous reactions in same setup using interchangeable tubular attachments
- Pressures upto 350 bar
- Temperatures upto 400°C
- Offered with gas flow controller & liquid metering pump
- Very compact, versatile & economical

FOR HOMOGENEOUS REACTIONS

- PTFE, SS 316 or Hastelloy C helical tubular reactor in 4, 8, 16 ml volume to fit in same furnace

APPLICATIONS

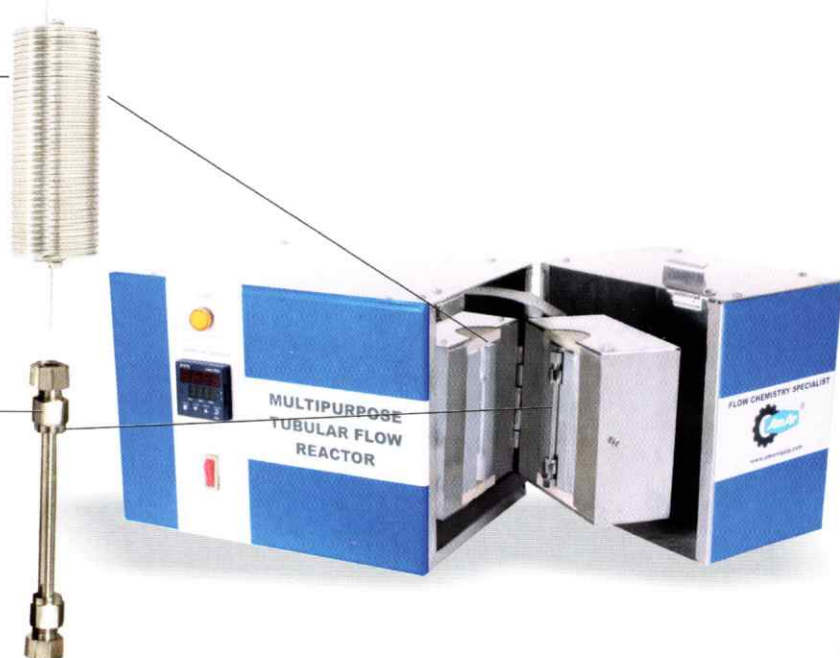
- Alkylation
- Heterocycle synthesis
- Supercritical reaction etc.

FOR HOMOGENEOUS REACTIONS

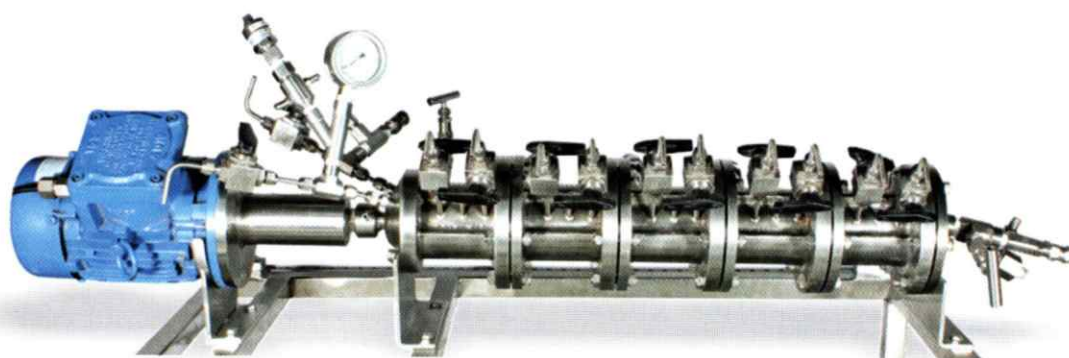
- SS316 or Hastelloy C vertical catalyst holding tube in 1, 2, 4 ml volume to fit in same furnace

APPLICATIONS

- Hydrogenation
- Oxidation
- Carboxylation etc.



SLURRY-FLO REACTOR



SALIENT FEATURES

- Multiple jacketed reaction vessels connected in series horizontally with common magnetically coupled agitator
- Offers a cost effective way of having high number of mixed stages
- It can handle clean fluids / immiscible fluids / slurry application or reactions with long reaction times
- Scalable to large volumes
- The operating pressures can be as high as 50 bar
- The reaction times could range from a few seconds to as high as 10 hours

TECHNICAL SPECIFICATIONS

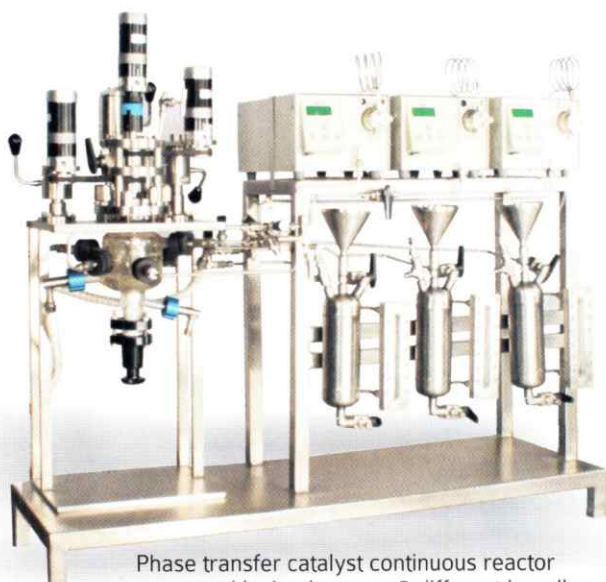
Volumes	250 ml to 5 ltr.
Flow rate (lph)	Upto 300
Design pressure (bar)	50
Design temperature (°C)	-30 to 200
Material of construction (MOC)	SS 316 / Hastelloy C 276
Heat transfer area (m ² /m ³)	60

CONTINUOUS STIRRED TANK REACTOR



SALIENT FEATURES

- Product is developed / produced on continuous basis for better productivity
- Stirred tank reactors of 25 ml to 10000 ltr
- Single or multiple reactors connected in series
- SS-316, Hastelloy C, inconel etc. materials
- Pressures upto 350 bar @ 500°C
- Ex-proof system for hazardous area
- Fully automated PC controlled systems to continuously monitor, record & control various parameters like temperature pressure, motor speed, gas / liquid flow etc.
- Gas mass flow controller, metering pumps, level controller, catalyst filtration system with SCADA software etc. are provided for a typical hydrogenation application.



Phase transfer catalyst continuous reactor system with simultaneous 3 different impeller speeds at 3 different levels



1 Ltr, 4 nos. CSTR in series



High pressure CSTR for hydrogenation



CSTR for hydro-cracking of heavy hydrocarbon oils



PACKED BED REACTORS

SALIENT FEATURES

- Fixed, fluidized, trickle bed designs offered.
- Reactor volumes from 10ml to 100 ltrs.
- Designed up to 350 bar & 1000°C.
- SS316, Inconel, Hastelloy C etc.
- Explosion proof plants for hazardous area.
- Tabletop or skid mounted plants.
- Customisable systems for gas liquid feed combinations, series or parallel reactors, multi zone heating furnaces etc. with integrated controls, high level of safety, automation and SCADA software.

APPLICATIONS

- Catalyst screening
- Hydrogenation
- Fischer-tropsch process
- Hydro-cracking
- Vapor phase reaction
- Coal to syngas production
- Pyrolysis reactions
- Biomass gasifications



Micro reactor assembly for
Liposome formation
Reactor volume: 100 ml
Pressure: 100 bar
Temperature: 500 °C
Material: Hastelloy C

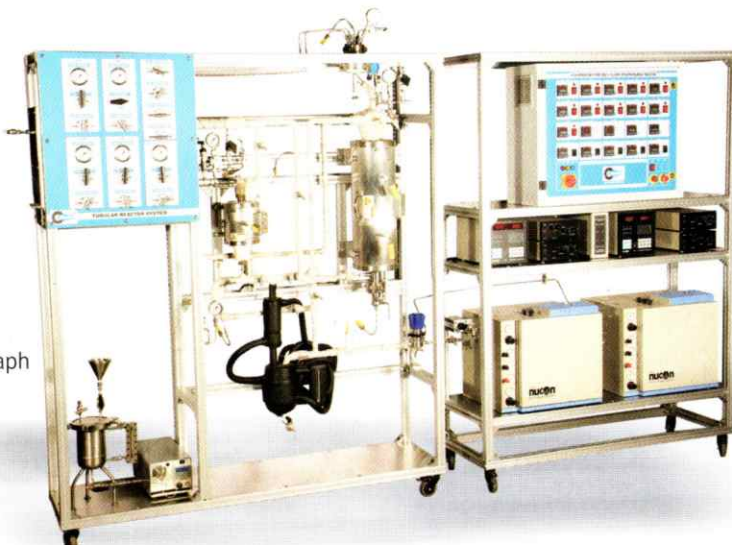


2 reactors with flexibility to connect
in parallel & series
Reactor volume: ~30 ml
Pressure: 5 bar.
Temperature: 1000 °C.
Materials: Inconel



Table Top
Reactor
System
(TTRS)

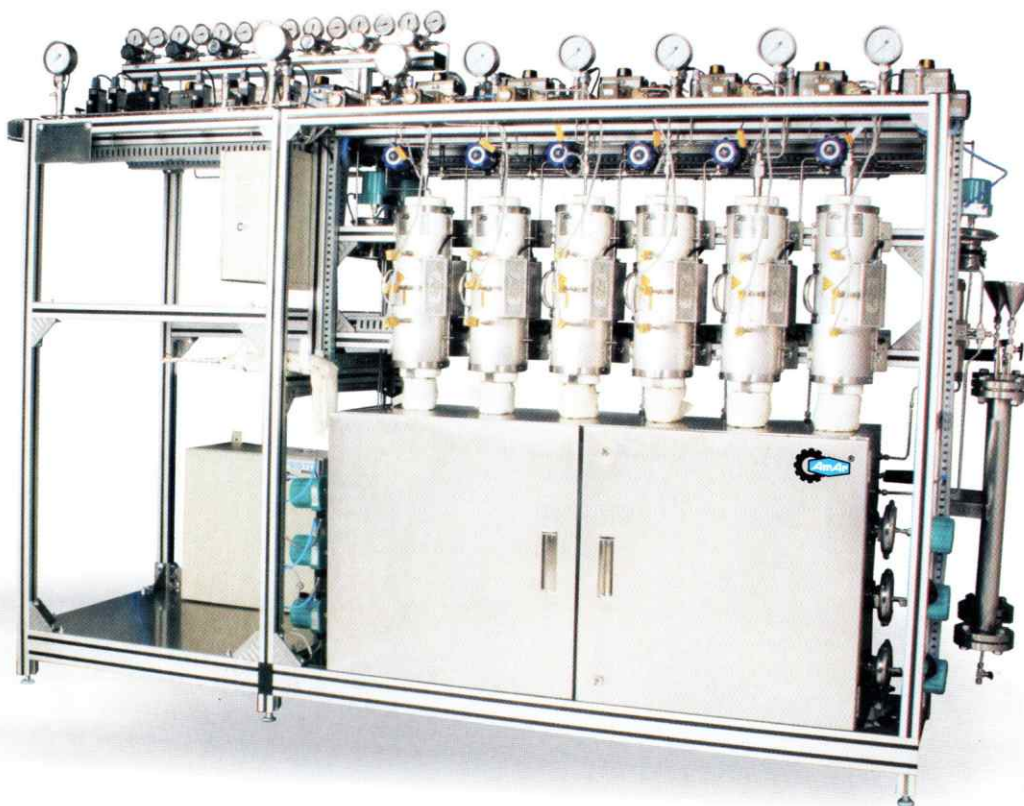
Reactors: 2 nos.
Pressure: 100 bar
Temperature: 700 °C
With gas chromatograph



PACKED BED REACTORS

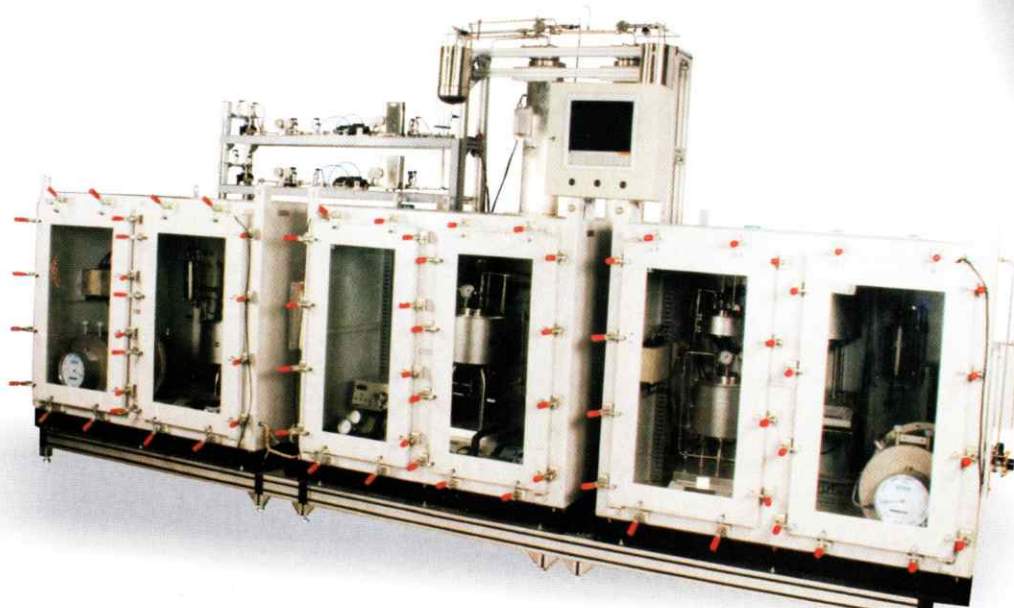
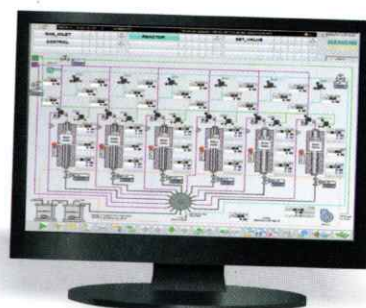


High throughput multi tube reactor system for catalyst screening



SYSTEM SPECIFICATIONS

- Reactor volume: 6 Nos- parallel- 10 to 100 ml. same or of different volumes (optional: upto 16 nos in parallel).
- Design pressure: 100 bar
- Design temperature: 900 °C
- Material: SS 316, Inconel, Hastelloy C, etc.



Pilot scale tubular reactor with purge panels for continuous biojet fuel production



OPTIONAL ACCESSORIES

GAS PRESSURE REGULATOR

To manually charge different gases at desired pressures upto 140 bar / 2000 psi or higher into the reactor from gas cylinder. Nitrogen, Oxygen & Hydrogen can be charged through same regulator (with special adaptor). The regulator is made from SS316 & comes with inlet - outlet pressure gauges & flexible SS braided PTFE high pressure hose pipe (4m long) with check valve.



THERMAL GAS MASS FLOW METER (MFM) / CONTROLLER (MFC)

MFM can be used to measure accurate mass flow rate of gas (in gm/hr or LPH) & totalized quantity of mass / volume (in gms/ltr) charged in the reactor at any point. Mass flow controller (MFC) is used to charge the set flow rate of gas into the reactor at high pressures upto 100 bar. The same MFM / MFC can be used for multiple gases by just entering the conversion factor for different gas densities provided the gases are inert to each other. The MFM/MFC comes with high pressure flexible hose, inlet filter with digital gas flow indicator.



MFC



Flow Indicator & Totaliser

CORIOLIS GAS - LIQUID MASS FLOW METER / CONTROLLER

These are used for higher & accurate gas or liquid flow rate indication or control in cases where thermal mass flow meters are not suitable. A common meter can be used for different gases & liquids for a particular range of flow.



Coriolis Mass Flow Meter



Coriolis Mass Flow Controller

DIGITAL PRESSURE INDICATOR

It consists of SS316 / Hastelloy C pressure sensor (transmitter) & digital pressure indicator/controller (mounted on common control panel) with pressure alarm.



Pressure Sensor

LIQUID METERING PUMP SYSTEM

This system is used to charge liquid at a desired rate from as low as 1 ml/hr to 100 ltr/hr, is under pressurized condition.

- High pressure accurate HPLC type low flow metering pumps for high pressures upto 350 bar & flow range from 0.01 upto 100ml / min. Materials: SS316, option: Hastelloy C, Titanium
- Diaphragm metering pumps for pressures upto 100 bar & minimum flow range of 60-600 ml/hr to maximum 10-100 lit/hr. The flow rates are varied by varying the motor speed with variable frequency drive. Materials: SS316, option: Hastelloy C, Titanium
- Low / high pressure syringe, pumps can be offered for pumping corrosive chemicals.



(a)



(b)



(c)

WEIGHING BALANCE

To measure precise amount / quantities of feed / product consumed or produced during process.



HEATING COOLING CIRCULATORS

- Single fluid closed loop system from -90 °C to 250°C.
- Heating cooling bath circulators from -70°C to 175°C.
- High temperature circulators from ambient to 350°C.
- Chillers up to -15°C.
- Suitable for reactor volume from 10 ml to 3000 ltr.



OPTIONAL ACCESSORIES



FLOW CONTROL VALVES

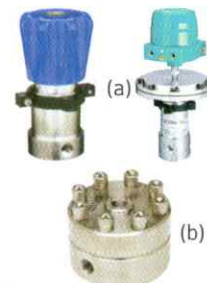
These valves are used to control flow, pressure or level



BACK PRESSURE REGULATOR

It is SS 316 regulator mounted on the vent line of the reactor & is used for maintaining constant pressure inside the reactor upto 350 bar. The pressure is maintained by releasing the excess pressure into the atmosphere.

- Optional:**
- a) Electronic actuated digital pneumatic back pressure regulator, (6 bar air supply is set digitally & can be released at preset rate of pressure release required).
 - b) Pneumatically actuated pilot operated back pressure regulator (air / N₂ gas supply for rated pressure is required)
 - c) Electronic control unit & forward regulator with 4 mtr. hose for activating (b) above.
 - d) Materials: Hastelloy C, PTFE etc.



PRESSURE SAFETY VALVE & SAFETY RUPTURE DISC

Safety rupture discs can be provided for pressure ratings of 100 bar & pressure safety valves can be provided for any pressures from 1 to 350 bar with provision to vary release pressure within a certain range. These valves come with PTFE / Viton / Kalrez 'O' rings.



Rupture Disc
Pressure Safety Valve

SCADA SOFTWARE FOR REMOTE OPERATION & RECORDING

SCADA is a supervisory control & data acquisition software with all controllers / indicators having RS 485 modbus communication port or PLC & HMI / touch panel, for online display, set point changes & data logging of various parameters like pressure, temperature, liquid / gas flow rate, oil/heater temperature, level, pH, ORP, IR etc. remotely from PC as well as locally from panel. It gives continuous online data logging at predefined (variable) time interval, online graphical representation as well as historical data & graphs on PC for single or multiple autoclaves. RS 485-232 convertor & cable upto 50 m or higher is also supplied.



PURGE CABINET

Cabinet with nitrogen purging for explosion proof zones can be offered by mounting nonexplosion proof instruments & assembly inside the cabinet



WET GAS METER

Wet gas meter is an analog instrument used to measure exhaust gas flow for gas mixer & is available in all flow rate ranges. It can give output for digital flow indication

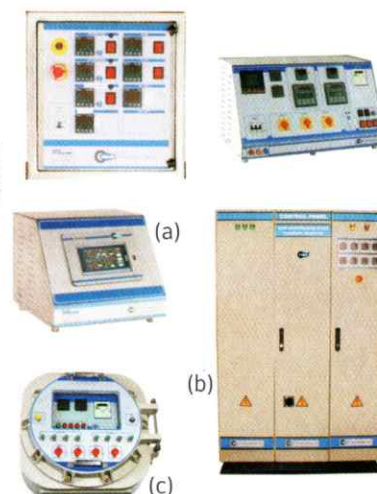


CONTROL PANEL

Control panel consists of programmable PID temperature controller cum indicator with temperature alarm system (settable), safety alarm & heater trip system for malfunctioning of controller / sensor/ temperature rise beyond set limit. Digital pressure indicator / controller, gas / liquid flow indicator, totaliser, level etc. indicators are provided additionally on same common control panel depending on the optional accessories selected.

Optional:

- a) Touch screen panel with SCADA software.
- b) PLC based control panel with touch panel HMI or remote SCADA software & PC control.
- c) Complete ex-proof - flame proof (FLP) Group IIA / IIB or IIC, ATEX zone 1, class 1 div. 2 certified.





FLOW CHEMISTRY SERVICES

**R&D facility dedicated to flow chemistry
feasibility trials & process development**



Flow chemistry

Process development

Piloting

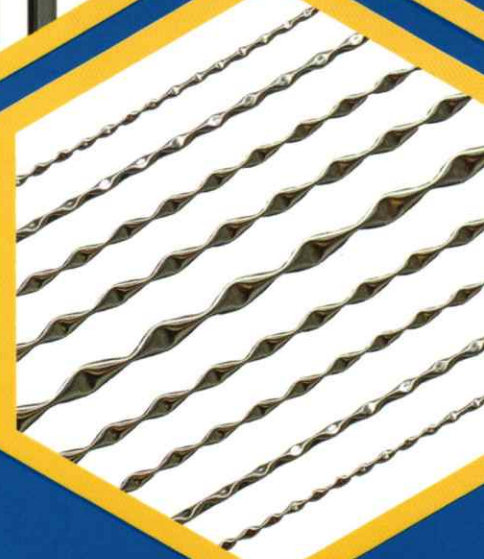
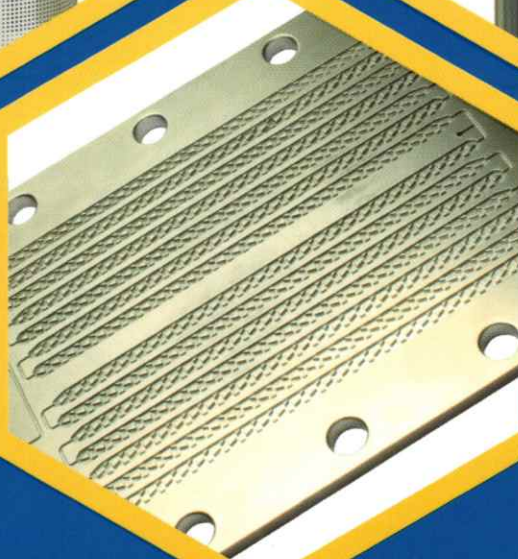
Custom equipment design

Commercialization support

High pressure chemistry

Custom synthesis





PRESSURE and FLOW Reactors

PRESSURE REACTOR & SYSTEMS



STIRRED PRESSURE AUTOCLAVES



- Volume: 100 ml – 100 ltr.
- Pressures up to 350 bar @ 500°C.
- SS316, Hastelloy, Monel, Inconel, Titanium, Zirconium.
- Zero leakage magnetic coupling.
- Ex-proof, CE / CSA certified electricals, PED / ASME, U stamp certified pressure reactors

PLANT SCALE PRESSURE REACTORS



- Volume: 100 ltr – 10000 ltr.
- Pressures up to 100 bar @ 300°C.
- Material: SS316, Hastelloy, Inconel, Titanium, Zirconium.
- High mass transfer gas induction impeller / PBT impeller.
- Zero leakage magnetic coupling.
- Ex-proof, CE / CSA certified electricals, PED / ASME, U stamp certified pressure reactors.

PARALLEL SYNTHESIZER



- Volume: 100 ml – 2 ltr.
- Pressures up to 350 bar @ 500°C.
- SS316, Hastelloy, Monel, Inconel, Titanium, Zirconium.
- 4 or 6 parallel reactors.
- Fully automated with SCADA software.

ECO CATALYST SCREENING



- Volume: 25 ml x 7 Nos.
- Pressures up to 100 bar @ 200°C.
- For high throughput catalyst screening.
- Bottom stirred parallel reactors..



PRESSURE REACTOR & SYSTEMS

GLASS AUTOCLAVES



- Volume: 100 ml – 2 ltr.
- Design pressure: Up to 10 bar.
- Design temperature: Up to 300°C.
- Material: SS316, Hastelloy, Glass.
- Interchangeable metal & glass autoclaves.

MAGNETIC DRIVE COUPLINGS / STIRRERS



- For reactor volume: 50 ml to 10000 ltr.
- Design pressure: full vacuum to 700 bar
- Material: SS316, Hastelloy C
- For high vacuum distillation in RBF
- Mixer with inline motor & coupling
- High torque coupling up to 1000 Nm

PRESSURE VESSELS



- Volume: 100 ml – 10000 ltr.
- Pressures up to 700 bar & temperature up to 600°C.
- SS316, Hastelloy, Monel, Inconel, Titanium, Zirconium.
- PED / ASME, U stamp certified pressure vessels.

ACID DIGESTION BOMB



- Volume: 50 ml to 750 ml.
- Pressures up to 200 bar @ 250°C.
- Material: SS-316 with PTFE liner & cap.
- Microwaveable acid digestion bombs.

SHAKER HYDROGENATOR



- Volume: 100 ml – 2 ltr.
- Design pressure: Up to 20 bar for metal vessels and 3 bar for glass vessels.
- Design temperature: Up to 200°C for metal vessels and up to 150 °C for glass vessels.
- Material: SS316, Hastelloy C

SUPERCritical FLUID EXTRACTION



- Volume: 100 ml – 1000 ltr.
- Design pressure: Up to 700 bar.
- Feed capacity of 0.24 to 2400 kg/day.
- For extraction of herbs, spices, essential oils, oleoresins, flavours, fragrance, colours.
- Decaffeination of tea & coffee.
- Textile dyeing of fabrics, drying of aerogels.
- Synthesis of nano particles

HPHT CORROSION TESTING



- Volume: 500 ml – 100 ltr.
- Design pressure: Up to 350 bar.
- Design temperature: Up to 600°C.
- Material: SS316, Hastelloy C, Inconel, Titanium
- For static, dynamic, recirculating loop & electrochemical corrosion studies.

GAS HYDRATE FORMATION SYSTEM



- Volume: 100 ml – 100 ltr.
- Pressures up to 350 bar @ 100°C.
- Glass window with camera to observe gas hydrate formation.
- Material: SS-316, Hastelloy, Inconel, Titanium.
- For analysis of effectiveness & efficiency of thermodynamic & kinetic gas hydrate inhibitors.



HIGH PRESSURE CUSTOMIZED SYSTEMS

Amar has expertise to design, manufacture & integrate customised high pressure reactor systems with all accessories, utilities, automation on a common skid for various applications.

100 ltr. Ex-proof hydrogenation pilot plant with catalyst filter & recycle, auto-cooling system, vacuum pump & catch pot.



10 ltr. Fully automated Ex-proof alkoxylation pilot plant



High pressure polytest reactor system



High pressure Ex-proof distillation skid



100 ltr. polymerization reactor for 2 million cP viscous polymer with hydraulic vessel raising, lowering tilting mechanism



Continuous emulsion polymerization pilot plant



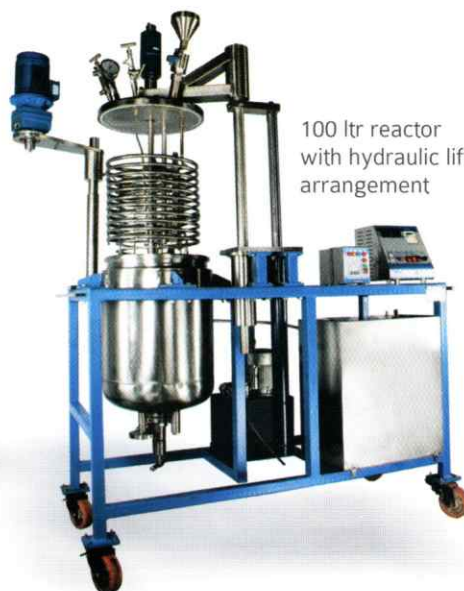
AUTOMATED & EX-PROOF PILOT PLANTS



Complete turn-key solutions can be offered for lab to plant scale with desired design, safety & hazardous area certification.



500 Ltr. Hydrogenation pilot plant with catalyst filter & recycle



100 Ltr reactor with hydraulic lifting arrangement



Ex-proof pilot plant for EO-PO reactions



Multipurpose pressure reactor with column distillation assembly



Fully automated & semi continuous pilot plant for styrene butadiene emulsion polymerization with pneumatic vessel raising, lowering & tilting

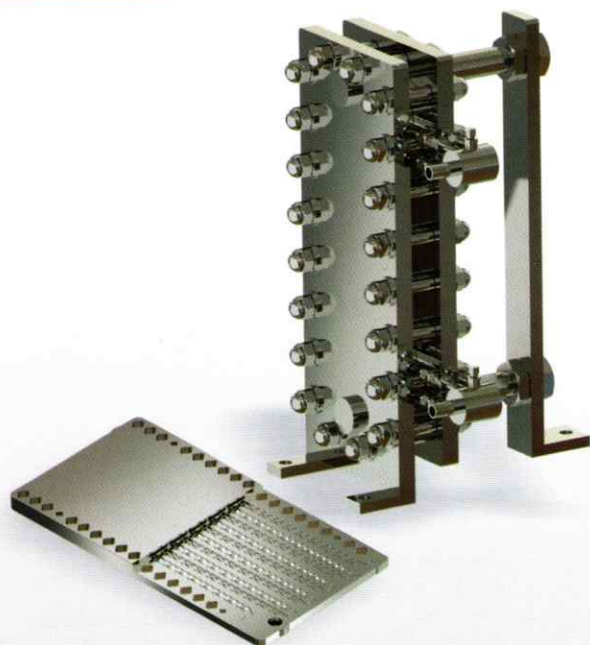
Ex-proof autoclave with load cell arrangement





FLOW REACTOR & SYSTEMS

AMaR-2: METAL MICROREACTOR



- Volume: 1 ml to 2 ltr.
- Flow rate: Up to 720 lph.
- Up to 20 bar & -50°C to 200°C.
- Material: SS316 / Hastelloy C.
- 3D printed microreactors.
- Ideal for fast liquid-liquid reactions with high exothermicity.

AMaR-3: CONICAL 3D FLOW REACTOR



- Volume :1 ml to 2 ltr.
- Flow rate: up to 720 lph.
- Up to 100 bar & -50°C to 350°C.
- Material: SS-316 / Hastelloy C.
- Ideal for fast & exothermic immiscible liquid-liquid and gas-liquid reactions.

AMaR-4P: PINCHED TUBE REACTOR



- Volume: 100 ml – 200 ltr.
- Flow rate: Up to 10,000 lph.
- Up to 50 bar & -50°C to 350°C.
- Material: SS316, Hastelloy C.
- Production scale versatile reactor suitable for various liquid-liquid or gas-liquid exothermic reactions, liquid-liquid extraction.

MULTI-FLO REACTOR



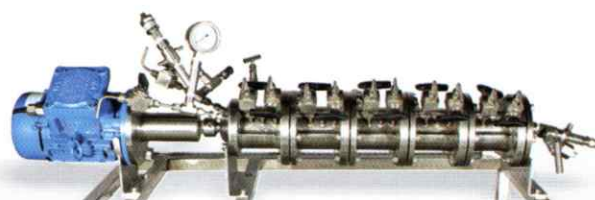
- Cost-effective starter set for proof-of-concept studies.
- Interchangeable reactor modules of AMaR-2, AMaR-3, AMaR-4, AMaR-4P with heating cooling.
- Up to 50 bar & -50°C to 350°C.
- Material: SS316, Hastelloy C.

TUBULAR REACTOR WITH STATIC MIXER



- Volume: 100 ml – 200 ltr.
- Flow rate: up to 10,000 lph.
- Up to 100 bar & -50°C to 350°C.
- Material: SS316, Hastelloy C.
- Optional static mixer elements for better mixing.
- Ideal for pilot to plant scale liquid-liquid, gas liquid & slurry applications.

SLURRY-FLO REACTOR



- Volume: 250 ml to 1 ltr.
- Up to 50 bar & -30°C to 300°C.
- Material: SS316, Hastelloy C.
- Multiple agitated cells connected in series.
- Each cell temperature control and sampling.
- Ideal for lab to pilot scale liquid slurry reactions and slow reactions with high residence times.

CONTINUOUS STIRRED TANK REACTOR



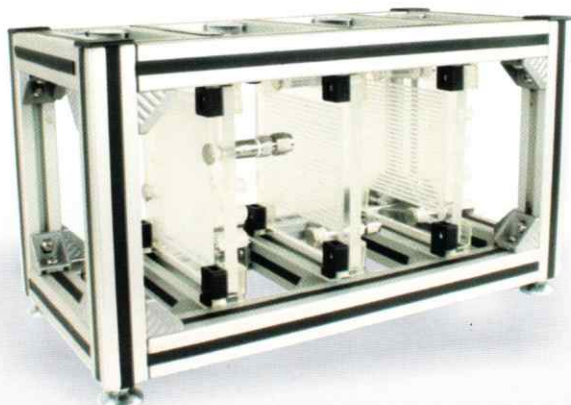
- Volume: 100 ml to 10000 ltr.
- Up to 350 bar @ 500°C.
- SS316, Hastelloy, Monel, Inconel, Titanium, Zirconium.
- Single or multiple reactors connected in series.
- Multiple inlets and outlets for addition & transfer / removal of gases & liquids.

PHOTO-FLO REACTOR



- Volume: 12 ml to 200 ml.
- Flow rate: Up to 150 lph.
- Up to 18 bar & -20°C to 150°C.
- Material: PFA.
- Jacketed PFA coil around light source UV lamp.
- Ideal for applications like photo halogenations, photo alkylation, vitamin D production, etc.

GLASS MICROREACTOR



- Volume: 0.1 ml to 50 ml.
- Up to 15 bar & 300 °C
- Material: Glass, quartz, silicon, glass-silicon compounds.
- Integrated multilayer glass construction for mixing, reaction & heat transfer.

SOLI-FLO REACTOR



- Homogeneous and heterogeneous reactions in same setup using interchangeable tubular attachments.
- Volume 1 ml to 16 ml.
- Up to 350 bar @ 500°C.
- Material: SS316, Hastelloy C.
- Ideal for gas-liquid-solid & vapor phase chemistries.

PACKED BED REACTOR



- Volume : 10 ml to 100 ltr
- Up to 350 bar & 1100°C
- For catalyst screening, oxidation, reforming, hydrogenation, liquefaction, fischer tropesch process, hydro-cracking, trickle bed reactions, vapor phase reactions, catalyst testing, fluid catalytic cracking, etc.

FLUIDIZED BED REACTOR



- Up to 12 bar & 1050°C
- Solid feed: Batch / continuous
- For syngas generation, pyrolysis reactions, biomass / coal gasification reactions, etc.

FLOW REACTOR & SYSTEMS



ELECTROCHEMICAL FLOW REACTOR



- Volume: 5 ml to 3000 ltr.
- Design pressure: Up to 2 bar & 120°C.
- Wetted part: PTFE.
- Graphite, lead, copper, platinum, aluminium, iron, titanium, Mg, SS, Zn, Cu/Ni electrodes.

LIQUID LIQUID SEPARATOR



- Flow rate: Up to 12 lph.
- Design temperature: Up to 120°C.
- Wetted part: PTFE.
- Advanced membrane-based liquid-liquid separation.
- Highly efficient than gravity based separations, even for challenging emulsions.

HEATING COOLING CIRCULATORS



- Single fluid closed loop system from -90 °C to 250°C.
- Heating cooling bath circulators from -70°C to 175°C.
- High temperature circulators from ambient to 350°C.
- Chillers up to -15°C.
- Suitable for reactor volume from 10 ml to 3000 ltr.

AGITATED NUTSCHE FILTER & DRYER



- 1 ltr – 100 ltr slurry volume.
- Design pressure: -1 to 4 bar.
- Design temperature : - 20°C to 150°C.
- Material: SS316, Hastelloy C.
- 0.005 m² to 0.5 m² filtration area.





AMAR EQUIPMENT PVT. LTD.

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